

SEMI-PARAMETRIC ESTIMATION OF INEQUALITY MEASURES

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Summary: Measures of inequality are very popular in economics, especially in measuring the inequality in income or wealth within a population and between populations. In this paper we propose semi-parametric estimators of measures of inequality, building on previous proposals for such estimators. Our proposals make explicit use of results from extreme value theory and are applied to the Gini coefficient, the generalised entropy and the quintile share ratio measures. The properties of these estimators are studied through a simulation, including their mean squared errors, sensitivity to outliers and confidence intervals based on them. They are found to perform very well compared to the traditional nonparametric estimators. The results are applied to an income data set. Some areas of future research are indicated.

1. Introduction

Measures of inequality, also used as measures of concentration or diversity, are very important in economics, especially in measuring income inequality. They also have applications in many other branches of science, e.g. in ecology (see e.g. Magurran, 1991), sociology (see e.g. Allison, 1978), demography (see e.g. White, 1986) and information science (see e.g. Rousseau, 1993). Over the years a large number of measures have been proposed, with the Gini index perhaps the most well-known one. Some other well-known measures are the generalised entropy class, the Atkinson class and the quintile share ratio. These measures have been studied extensively and their properties investigated in many papers. Such papers included the study of their strong and weak points and their comparative performance with respect to axiomatic properties. The measures have also been used extensively in practice, especially in studying income inequality. The Gini coefficient has been

applied more recently in modern financial portfolio analysis. The quintile share ratio has become more well-known in recent years due to their inclusion by the European Union as part of its indicators of poverty and social exclusion. For a good overview of inequality and its measurement see e.g. Cowell (2000). For the application of the Gini coefficient in finance, see e.g. Shalit and Yitzhaki (2005). For the quintile share ratio see e.g. EU-SILC 131/04 (2004), Hullinger and Schoch (2009) and Langel and Tillé (2011).

Having reliable inequality measures available is an important first step. A next step is to estimate the values of these measures using samples from the appropriate populations and, in particular, to estimate the variability of these estimators and more generally, to obtain confidence intervals for the measures. Since inequality is inherently dependent on the tails of a population, estimators of inequality are typically sensitive to data from these tails. It is well-known that income distributions often exhibit a long tail to the right, making estimators of inequality particularly sensitive to large values. It is thus important to study the behaviour of estimators based on data from heavy tailed distributions. Since measures of inequality are functionals of the underlying distribution function, a reliable estimator for a measure therefore depends critically on a reliable estimator for the distribution function. In many traditional estimators the distribution function is estimated by the empirical distribution function and the latter substituted into the corresponding functional to obtain a plug-in estimator. However, the empirical distribution function is not a reliable estimator for long tailed distributions and in such cases the corresponding estimator for the measure of inequality is not reliable and remedial action must be taken. This remedial action can be either a trimming of the extreme data (see e.g. Cowell and Victoria-Feser, 2006) or a modification of the distribution function estimator to make it more robust in the case of long tailed distributions.

Cowell and Flachaire (2007) have proposed a so-called semi-parametric approach to modify estimators under heavy tailed distributions. In a semi-parametric approach models are used that are partly parametric and partly nonparametric. They are useful in many areas of statistical modelling – (see e.g. Fernandez-Sainz and Rodrigues-Poo, 2010; Nascimento, Gamerman and Lopes, 2012; Gomes, De Haan and Peng., 2002). In the Cowell-Flachaire semi-parametric approach they estimate the left part of the distribution, where the bulk of the data resides, using the usual nonparametric empirical distribution function and the right (upper) part of the distribution using a Pareto distribution. The resulting estimator is therefore partly nonparametric and partly parametric, hence semi-parametric. Based on this approach in Cowell and Flachaire (2007), we further develop the semi-parametric estimation procedures using results from Extreme Value Theory (EVT) to obtain more reliable distribution estimators. Substituting these semi-parametric estimators of the distribution function into the functional corresponding to an inequality measure, we obtain more robust estimators of these measures. We particularly focus on the Gini, the generalised entropy and the quintile share ratio measures, for which we construct semi-parametric estimators and we study some of their properties and their sensitivity to contaminations.

The layout of the paper is as follows. In the next section the approach to semi-parametric estimation of the underlying distribution function is discussed and in Section 3 we give the semi-parametric estimators for the measures considered, including a method of deciding between models. A simulation is carried out in Section 4 to study some properties of these estimators, including their mean squared errors, their sensitivity to contaminations and confidence intervals based on them. We also compare the performance of the procedures developed in this paper to the traditional nonparametric

methods. An illustration is given in Section 5 to show the application of the proposed methods. We conclude in Section 6, where we also give a number of further aspects that can be investigated in future research.

Throughout this paper we will assume that the random variable of interest X is positive having continuous distribution function F . Denote by $X_1, X_2, \dots, X_n$ a sample from F and by $X_{1,n} < X_{2,n} < \dots < X_{n,n}$ the corresponding order statistics.

2. Distribution function estimators based on extreme value theory

In this section we define a semi-parametric estimator for the underlying distribution function, specifically applicable to heavy tailed distributions. This semi-parametric estimator is constructed by combining the usual nonparametric estimator in the bulk of the data with a parametric estimator for the tails. In our approach we make use of results from extreme value theory (EVT) to estimate the parametric component.

In order to construct such a semi-parametric estimator, first define a semi-parametric distribution function by

$$F(x) = \begin{cases} F(x), & x \leq x_0 \\ F(x_0) + (1 - F(x_0))F_\theta(x), & x > x_0 \end{cases} \quad (1)$$

for a given x_0 , where F_θ is a parametric distribution defined on (x_0, ∞) and satisfying the condition $F_\theta(x_0) = 0$. Note that θ is typically vector valued. Here F is the unknown underlying distribution function that will be estimated (in the bulk of the data) by the empirical distribution function. Taking x_0 as an upper quantile of F , say $x_0 = Q(1 - \alpha; F)$, leads to

$$F(x) = \begin{cases} F(x), & x \leq Q(1 - \alpha; F) \\ 1 - \alpha(1 - F_\theta(x)), & x > Q(1 - \alpha; F). \end{cases} \quad (2)$$

In (7), estimate the component F by F_n , the empirical distribution function and choose $\alpha = \frac{k}{n}$, for some k . Then $X_{n-k,n}$ can be taken as an estimator for $Q(1 - \alpha; F)$. This together with $\hat{\theta}$, an estimator for θ , leads to a semi-parametric estimator for the underlying distribution function given by

$$\tilde{F}_n(x) = \begin{cases} F_n(x), & x \leq X_{n-k,n} \\ 1 - \frac{k}{n}(1 - F_{\hat{\theta}}(x)), & x > X_{n-k,n}. \end{cases} \quad (3)$$

For the choice of F_θ in this estimator, we will make use of results from EVT.

2.1. Results from extreme value theory

As mentioned above, EVT is used to estimate the parametric part in (7). Denote by u an appropriate threshold and consider the conditional distribution of the exceedance of u given that u was exceeded. Taking F to belong to the domain of attraction of the generalised extreme value distribution, the following limiting results hold as $u \rightarrow \infty$ (see Beirlant, Goegebeur, Segers and Teugels, 2004).

1. The distribution of the absolute exceedance $X - u$ given $X > u$ converges to the generalised Pareto distribution (GPD) defined by

$$G(x; \sigma, \gamma) = 1 - (1 + \sigma^{-1}\gamma x)^{-\gamma^{-1}}, x > 0,$$

with $\sigma > 0$ a scale parameter and $\gamma > 0$ a shape parameter, called the extreme value index (EVI).

2. The distribution of the relative exceedance X/u given $X > u$ converges to the strict Pareto distribution defined by

$$G(x; \gamma) = 1 - x^{-\gamma^{-1}}, x > 1, \text{ with } \gamma > 0 \text{ the extreme value index.}$$

3. The departure from Pareto using a second order refinement of the result in (1) leads to an approximation of the distribution of the relative exceedance by a mixture of two Pareto distributions, called the perturbed Pareto distribution (PPD), having survival function

$$\bar{G}(x; \gamma, c, \tau) = (1 - c)x^{-\gamma^{-1}} + cx^{-\gamma^{-1}-\tau}, x > 1, \gamma > 0, \tau > 0, c \in (-\tau^{-1}, 1).$$

2.2. Semi-parametric estimators for the distribution function

Based on the EVT results discussed above, a semi-parametric estimator for the underlying distribution function is obtained in each of the three cases and these are then used to estimate the inequality measures. A convenient choice of threshold is a large order statistic, say $u = X_{n-k,n}$, leading to exactly k exceedances. Using $X_{n-k,n}$ as threshold then leads to the following three estimators for F based on results (1), (2) and (3). In these, take n large and k relatively small so that $X_{n-k,n}$ is large.

Using the approximation in (1) leads to the estimator

$$\tilde{F}_n(x) = \begin{cases} F_n(x) & \text{for } x \leq X_{n-k,n} \\ 1 - \frac{k}{n} \left(1 + \frac{\hat{\gamma}(x - X_{n-k,n})}{\hat{\sigma}} \right)^{-\hat{\gamma}^{-1}} & \text{for } x > X_{n-k,n}. \end{cases} \quad (4)$$

Using the approximation in (2) leads to the estimator

$$\tilde{F}_n(x) = \begin{cases} F_n(x) & \text{for } x \leq X_{n-k,n} \\ 1 - \frac{k}{n} X_{n-k,n}^{\hat{\gamma}^{-1}} x^{-\hat{\gamma}^{-1}} & \text{for } x > X_{n-k,n}. \end{cases} \quad (5)$$

Using the approximation in (3) leads to the estimator

$$\tilde{F}_n(x) = \begin{cases} F_n(x) & \text{for } x \leq X_{n-k,n} \\ 1 - \frac{k}{n} X_{n-k,n}^{\hat{\gamma}^{-1}} \left[(1 - \hat{c}) x^{-\hat{\gamma}^{-1}} + \hat{c} X_{n-k,n}^{\hat{\tau}} x^{-\hat{\gamma}^{-1}-\hat{\tau}} \right] & \text{for } x > X_{n-k,n}. \end{cases} \quad (6)$$

Remark. Although we investigated a number of estimation methods for the parameters, in our applications, $\hat{\gamma}$, $\hat{\sigma}$, $\hat{\tau}$ and $\hat{c}$ will be taken as the maximum likelihood estimators based on the exceedances. For our purposes their performance and ease of computation were found to be quite satisfactory.

3. Semi-parametric estimators for measures of inequality

In this section we first define measures of inequality, and then we use the semi-parametric estimators of the underlying distribution developed in the previous section to derive semi-parametric estimators for these measures. We particularly consider the Gini coefficient, the generalised entropy and the quintile share ratio.

- The Gini index defined by

$$I_G = \mu_1^{-1} \int_0^\infty F(x)(1 - F(x)) dx.$$

Remark. There are many alternative definitions of the Gini index available in the literature. We used this definition since it is convenient for computational purposes.

- The generalised entropy measure defined by

$$I_E^\alpha = [\alpha(\alpha - 1)]^{-1} \int_0^\infty \left[\left(\frac{x}{\mu_1} \right)^\alpha - 1 \right] dF(x) = [\alpha(\alpha - 1)]^{-1} \left(\frac{\mu_\alpha}{\mu_1^\alpha} - 1 \right),$$

where $\mu_\alpha = \int_0^\infty x^\alpha dF(x)$, $\alpha \in \mathbb{R}$. The two special cases $\alpha = 0$ and $\alpha = 1$ correspond respectively to the mean logarithmic deviation given by

$$I_E^0 = \int_0^\infty \log \left(\frac{x}{\mu_1} \right) dF(x) = \log \mu_1 - \int_0^\infty \log x dF(x),$$

and the Theil index given by

$$I_E^1 = \frac{1}{\mu_1} \int_0^\infty x \log x dF(x) - \log \mu_1.$$

- The quintile share ratio defined by

$$I_Q = \frac{\int_0^\infty \mathbb{I}_{Q(0.8;F)}(x) dF(x)}{\int_0^\infty \mathbb{I}_{Q(0.2;F)}(x) dF(x)}.$$

A thorough discussion of these measures can be found in e.g. Cowell and Flachaire (2007), Hullinger and Schoch (2009) and González, Velasco, Gavilán and Sánchez-Reyes (2010).

Denoting a measure of inequality (e.g. any of the previous measures) by $I(F)$, a semi-parametric estimator for $I(F)$ is then given by the plug-in estimator $I(\tilde{F}_n)$, where $\tilde{F}_n$ is one of the estimators in the previous section. We now give the form of $I(\tilde{F}_n)$ for the three cases considered, i.e. when fitting the GPD, the strict Pareto or the PPD in the tail. The expressions follow in a fairly straightforward fashion from the respective definitions, i.e. for each measure of inequality defined above, we substitute $\tilde{F}_n$ for F and we calculate the corresponding quantities using straightforward algebra. Details of this can be found in Kpanzou (2011).

3.1. Estimators based on GPD

Fitting the generalised Pareto distribution in the tail, leads in the case of the Gini coefficient to the semi-parametric estimator

$$I_{SPG} = (n\hat{\mu})^{-1} \sum_{i=1}^{n-k-1} i \left(1 - \frac{i}{n}\right) (X_{i+1,n} - X_{i,n}) + k\hat{\sigma} [2n - k - \hat{\gamma}(n - k)] / [n^2 \hat{\mu} (1 - \hat{\gamma})(2 - \hat{\gamma})], \quad (7)$$

with

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n-k} X_{i,n} + \frac{k}{n} \left(X_{n-k,n} + \frac{\hat{\sigma}}{1 - \hat{\gamma}} \right), \quad \hat{\gamma} < 1.$$

For the GE measures, the semi-parametric estimator is given by

$$I_{SPE}^{\alpha} = [\alpha(\alpha - 1)]^{-1} \left[\frac{\hat{\mu}_{\alpha}}{\hat{\mu}_1^{\alpha}} - 1 \right], \quad (8)$$

with the special cases $\alpha = 0$ and $\alpha = 1$ treated in the obvious fashion, and

$$\hat{\mu}_{\alpha} = \frac{1}{n} \sum_{i=1}^{n-k} X_{i,n}^{\alpha} + \frac{k\hat{I}_{\alpha}}{n\hat{\sigma}},$$

with

$$\hat{I}_{\alpha} = \int_{X_{n-k,n}}^{\infty} x^{\alpha} \left[1 + \frac{\hat{\gamma}(x - X_{n-k,n})}{\hat{\sigma}} \right]^{-\frac{1}{\hat{\gamma}} - 1} dx.$$

In the case of the quintile share ratio we obtain as semi-parametric estimator

$$I_{SPQ} = \begin{cases} 0.2n \left[X_{n-k,n} + \frac{\hat{\sigma}}{\hat{\gamma}} \left(\frac{1}{1 - \hat{\gamma}} \left(\frac{0.2n}{k} \right)^{-\hat{\gamma}} - 1 \right) \right] / \left[\sum_{i=1}^{[0.2n]} X_{i,n} \right], & n - k \leq [0.8n] \\ \left[\sum_{i=[0.8n]}^{n-k} X_{i,n} + k \left(X_{n-k,n} + \frac{\hat{\sigma}}{1 - \hat{\gamma}} \right) \right] / \left[\sum_{i=1}^{[0.2n]} X_{i,n} \right], & n - k > [0.8n]. \end{cases} \quad (9)$$

3.2. Estimators based on strict Pareto distribution

Fitting the strict Pareto distribution in the tail, leads in the case of the Gini coefficient to the semi-parametric estimator

$$I_{SPG} = (n\hat{\mu})^{-1} \sum_{i=1}^{n-k-1} i \left(1 - \frac{i}{n}\right) (X_{i+1,n} - X_{i,n}) + k\hat{\gamma}X_{n-k,n} [2n - k - \hat{\gamma}(n - k)] / [n^2 \hat{\mu} (1 - \hat{\gamma})(2 - \hat{\gamma})], \quad (10)$$

with

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n-k} X_{i,n} + \frac{kX_{n-k,n}}{n(1 - \hat{\gamma})}, \quad \hat{\gamma} < 1.$$

For the GE measures, the semi-parametric estimator is given by

$$I_{SPE}^{\alpha} = [\alpha(\alpha - 1)]^{-1} \left[\frac{\hat{\mu}_{\alpha}}{\hat{\mu}_1^{\alpha}} - 1 \right], \quad (11)$$

with the special cases $\alpha = 0$ and $\alpha = 1$ treated in the obvious fashion and

$$\hat{\mu}_\alpha = \frac{1}{n} \sum_{i=1}^{n-k} X_{i,n}^\alpha + \frac{kX_{n-k,n}^\alpha}{n(1-\alpha\hat{\gamma})}, \quad \hat{\gamma} < 1.$$

In the case of the QSR we obtain as semi-parametric estimator

$$I_{SPQ} = \begin{cases} \left[\frac{k}{1-\hat{\gamma}} \left(\frac{0.2n}{k} \right)^{1-\hat{\gamma}} X_{n-k,n} \right] / \left[\sum_{i=1}^{[0.2n]} X_{i,n} \right], & n-k \leq [0.8n] \\ \left[\sum_{i=[0.8n]}^{n-k} X_{i,n} + k \frac{kX_{n-k,n}}{1-\hat{\gamma}} \right] / \left[\sum_{i=1}^{[0.2n]} X_{i,n} \right], & n-k > [0.8n]. \end{cases} \quad (12)$$

3.3. Estimators based on PPD

Fitting the perturbed Pareto distribution in the tail, leads in the case of the Gini coefficient to the semi-parametric estimator

$$I_{SPG} = (n\hat{\mu})^{-1} \sum_{i=1}^{n-k-1} i \left(1 - \frac{i}{n} \right) (X_{i+1,n} - X_{i,n}) + \frac{k\hat{\gamma}X_{n-k,n}}{n\hat{\mu}} \left[\frac{1-\hat{c}}{1-\hat{\gamma}} + \frac{\hat{c}}{1+\hat{\gamma}(\hat{\tau}-1)} \right] \\ + \frac{k^2\hat{\gamma}X_{n-k,n}}{n^2\hat{\mu}} \left[\frac{(1-\hat{c})^2}{2-\hat{\gamma}} + \frac{\hat{c}^2}{2+\hat{\gamma}(2\hat{\tau}-1)} + \frac{2\hat{c}(1-\hat{c})}{2+\hat{\gamma}(\hat{\tau}-1)} \right], \quad (13)$$

with

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n-k} X_{i,n} + \frac{kX_{n-k,n}}{n} \left[\frac{1-\hat{c}}{1-\hat{\gamma}} + \frac{\hat{c}(1+\hat{\gamma}\hat{\tau})}{1+\hat{\gamma}(\hat{\tau}-1)} \right].$$

For the GE measures, the semi-parametric estimator is given by

$$I_{SPE}^\alpha = [\alpha(\alpha-1)]^{-1} \left[\frac{\hat{\mu}_\alpha}{\hat{\mu}_1^\alpha} - 1 \right], \quad (14)$$

with the special cases $\alpha = 0$ and $\alpha = 1$ treated in the obvious fashion and

$$\hat{\mu}_\alpha = \frac{1}{n} \sum_{i=1}^{n-k} X_{i,n}^\alpha + \frac{kX_{n-k,n}^\alpha}{n} \left[\frac{1-\hat{c}}{1-\hat{\gamma}\alpha} + \frac{\hat{c}(1+\hat{\gamma}\hat{\tau})}{1+\hat{\gamma}(\hat{\tau}-\alpha)} \right].$$

In the case of the QSR we obtain as semi-parametric estimator

$$I_{SPQ} = \begin{cases} \left[\frac{(1-\hat{c})}{1-\hat{\gamma}} X_{n-k,n}^{1/\hat{\gamma}} \hat{Q}(0.8)^{-1/\hat{\gamma}+1} + \frac{k\hat{c}(1+\hat{\gamma}\hat{\tau})}{1+\hat{\gamma}(\hat{\tau}-1)} X_{n-k,n}^{1/\hat{\gamma}+\hat{\tau}} \hat{Q}(0.8)^{-1/\hat{\gamma}-\hat{\tau}+1} \right] / \left[\sum_{i=1}^{[0.2n]} X_{i,n} \right], & n-k \leq [0.8n] \\ \left[\sum_{i=[0.8n]}^{n-k} X_{i,n} + kX_{n-k,n} \left(\frac{1-\hat{c}}{1-\hat{\gamma}} + \frac{\hat{c}(1+\hat{\gamma}\hat{\tau})}{1+\hat{\gamma}(\hat{\tau}-1)} \right) \right] / \left[\sum_{i=1}^{[0.2n]} X_{i,n} \right], & n-k > [0.8n] \end{cases} \quad (15)$$

where $\hat{Q}(\alpha)$ is a solution to the equation

$$(1 - \hat{c})x^{-1/\hat{\gamma}} + \hat{c}X_{n-k,n}^{\hat{\tau}}x^{-1/\hat{\gamma}-\hat{\tau}} = \frac{n}{k}X_{n-k,n}^{-1/\hat{\gamma}}(1 - \alpha).$$

Remark. The above expressions are easy to calculate for specific data sets. In the next two sections they will be applied to simulated data sets and to a real data set. Although the choice of k is unspecified here, many different procedures have been proposed in the literature for choosing k . In the simulation below we give a brief discussion of our choice.

3.4. Model selection

As seen above, we have three possible parametric distributions to fit in the tails. The question therefore is which of the three parametric distributions should be considered in an application. None of them can be said to be overall “best”, and so we need a further procedure to select the most appropriate one. Many measures exist to select the most appropriate distribution. Examples such as Kolmogorov-Smirnov and Anderson-Darling are typically goodness-of-fit based. Another closely related measure that is very easy to compute, is the so called representativeness index, proposed by Bertino (2006). This is defined as

$$R(\underline{X}, F) = 1 - \frac{12n}{4n^2 - 1} \sum_{i=1}^n \left(F(X_{i,n}) - \frac{2i-1}{2n} \right)^2. \quad (16)$$

This index is used to measure how well the sample $\underline{X}$ represents its parent distribution. Therefore, it can be used to select the model one should use when having to choose between different models. In this measure one would select the distribution giving the highest value. We will apply this in our data application in Section 5.

4. Simulation

In this section a simulation study is reported on that was carried out in order to determine the performance of the proposed SP estimators compared to the traditional nonparametric estimator, which is the plug-in estimator $I(F_n)$, with F_n the empirical distribution function. Two criteria of performance were considered, namely mean squared error and relative impact of contamination (discussed below). The performance of bootstrap confidence intervals based on SP estimators is also compared to the performance of those based on the nonparametric estimator. This is done in terms of the average confidence interval length (ACIL) and the coverage probability (CP).

Seven heavy tailed distributions were included, namely

- * The Pareto distribution with $x_0 = 0.1$ and $\beta = 1.5$ (denoted by Pa);
- * The Burr distribution with $\alpha = 2$, $\tau = 0.83$ and $\lambda = 1$ (denoted by Bu1);
- * The Burr distribution with $\alpha = 1$, $\tau = 1.4$ and $\lambda = 1$ (denoted by Bu2);

- * The Burr distribution with $\alpha = 0.5$, $\tau = 4$ and $\lambda = 1$ (denoted by Bu3);
- * The Fréchet distribution with $\alpha = 2$ (denoted by Fr1);
- * The Fréchet distribution with $\alpha = 1.7$ (denoted by Fr2);
- * The $|t_2|$ distribution (denoted by T2).

Four sample sizes were used viz. 500, 1000, 5000 and 10000, and the simulation was run over 1000 repetitions. The choice of a threshold is needed in two places, namely for estimating the distribution separation point as well as for estimating the parameters of the fitted parametric distribution.

A simulation was carried out over the above range of distributions and sample sizes and optimal values of k derived from the mean squared errors for each combination. From these an average, compromise choice of $k_{est} = h_{est}n$, $h_{est} \in [0, 1]$ was obtained for the estimation of the unknown parameters in the parametric distributions, which in turn were applied to $k_{SP} = h_{SP}\sqrt{n}$, $h_{SP} \in [0, 1]$. Different values of h_{est} and h_{SP} were obtained for each underlying distribution and each sample size. For example in the case of the Pareto distribution, we obtained on average $h_{est} = 0.1$ for all sample sizes, $h_{SP} = 0.3$ for $n = 500$, and $h_{SP} = 0.4$ for $n = 1000, 5000, 10000$. Although these compromise choices of k_{est} and k_{SP} gave satisfactory results, further research on this could lead to even further improvement of the performance of the estimators.

The estimators are denoted by SPGPD (when fitting the GPD in the tail), SPPa (when fitting the strict Pareto), and SPPPD (when fitting the PPD). The traditional nonparametric estimator is denoted by NP. The bootstrap percentile interval is denoted by BPI and the bootstrap t interval by BTI. See e.g. Efron and Tibshirani (1993) for a discussion of these methods. The simulation results are given in the tables below, where standard errors of the estimates are given in brackets. In the cases of MSE and RIC, the values in bold in each row are associated with the “best” estimates. For the confidence intervals, a number of coverage probabilities and average confidence interval lengths are highlighted in bold to show the better performance of the corresponding intervals.

4.1. Mean squared errors

The results in Table 1 are the mean squared errors (MSE) for estimators of Gini coefficient. These results indicate that for large samples ($n \geq 1000$), all estimators perform quite well over all the measures, but when the sample size is “small” ($n = 500$) the mean squared error becomes slightly larger due to bias in some cases and increased variance in others. For most of the cases, SPGPD, SPPa and SPPPD perform very well. The NP estimators become more competitive for large samples. Similar conclusions also hold for the other measures considered.

4.2. Relative impact of contamination

Denote by $\hat{F}$ an estimator for F using the original sample and by $\hat{F}^*$ the corresponding estimator using a contaminated sample. There are many ways of measuring the influence of contamination in the data on the estimators. There are also different ways of introducing data contamination. We

Table 1: Mean squared errors (MSE) for Gini Measure.

Gini (n=500)					Gini (n=1000)				
	NP	SPGPD	SPPa	SPPPD		NP	SPGPD	SPPa	SPPPD
Pa	0.0049 (0.0003)	0.0052 (0.0007)	0.0017 (0.0001)	0.0028 (0.0006)	Pa	0.0045 (0.0003)	0.0045 (0.0002)	0.0033 (0.0001)	0.0011 (0.0001)
Bu1	0.0019 (0.0001)	0.0025 (0.0003)	0.0580 (0.0086)	0.0016 (0.0003)	Bu1	0.0017 (0.0001)	0.0019 (0.0002)	0.0423 (0.0025)	0.0041 (0.0003)
Bu2	0.0044 (0.0002)	0.0081 (0.001)	0.0578 (0.0078)	0.0063 (0.0003)	Bu2	0.0033 (0.0002)	0.0058 (0.0003)	0.0393 (0.0011)	0.0045 (0.0000)
Bu3	0.0016 (0.0001)	0.0010 (0.0009)	0.0014 (0.0001)	0.0009 (0.0001)	Bu3	0.0014 (0.0001)	0.0006 (0.0001)	0.0010 (0.0000)	0.0005 (0.0000)
Fr1	0.0015 (0.0009)	0.0011 (0.0001)	0.0015 (0.0001)	0.0011 (0.0001)	Fr1	0.0011 (0.0006)	0.0006 (0.0001)	0.0011 (0.0000)	0.0004 (0.0001)
Fr2	0.0028 (0.0011)	0.0023 (0.0003)	0.0054 (0.0001)	0.0032 (0.0002)	Fr2	0.0021 (0.0009)	0.0015 (0.0000)	0.0040 (0.0000)	0.0014 (0.0002)
T2	0.0010 (0.0005)	0.0009 (0.0001)	0.0025 (0.0001)	0.0017 (0.0001)	T2	0.0008 (0.0002)	0.0005 (0.0000)	0.0017 (0.0000)	0.0005 (0.0001)
Average	0.0026 (0.0002)	0.0030 (0.0002)	0.0183 (0.0017)	0.0025 (0.0001)	Average	0.0021 (0.0002)	0.0022 (0.0001)	0.0132 (0.0004)	0.0018 (0.0001)

Gini (n=5000)					Gini (n=10000)				
	NP	SPGPD	SPPa	SPPPD		NP	SPGPD	SPPa	SPPPD
Pa	0.0017 (0.0000)	0.0013 (0.0001)	0.0036 (0.0000)	0.0002 (0.0000)	Pa	0.0017 (0.0001)	0.0014 (0.0000)	0.0043 (0.0000)	0.0003 (0.0000)
Bu1	0.0004 (0.0000)	0.0005 (0.0001)	0.0206 (0.0003)	0.0019 (0.0001)	Bu1	0.0002 (0.0000)	0.0004 (0.0000)	0.0161 (0.0001)	0.0017 (0.0001)
Bu2	0.0011 (0.0001)	0.0020 (0.0002)	0.0225 (0.0003)	0.0011 (0.0001)	Bu2	0.0010 (0.0001)	0.0014 (0.0000)	0.0184 (0.0001)	0.0007 (0.0001)
Bu3	0.0002 (0.0000)	0.0002 (0.0000)	0.0004 (0.0000)	0.0001 (0.0000)	Bu3	0.0001 (0.0000)	0.0001 (0.0000)	0.0003 (0.0000)	0.0001 (0.0000)
Fr1	0.0002 (0.0005)	0.0006 (0.0000)	0.0005 (0.0000)	0.0001 (0.0000)	Fr1	0.0002 (0.0004)	0.0001 (0.0000)	0.0003 (0.0000)	0.0001 (0.0000)
Fr2	0.0006 (0.0006)	0.0006 (0.0000)	0.0021 (0.0000)	0.0003 (0.0000)	Fr2	0.0005 (0.0003)	0.0004 (0.0000)	0.0015 (0.0000)	0.0001 (0.0000)
T2	0.0006 (0.0002)	0.0002 (0.0000)	0.0008 (0.0000)	0.0001 (0.0000)	T2	0.0001 (0.0001)	0.0001 (0.0000)	0.0006 (0.0000)	0.0001 (0.0000)
Average	0.0007 (0.0001)	0.0008 (0.0000)	0.0072 (0.0000)	0.0005 (0.0000)	Average	0.0005 (0.0001)	0.0006 (0.0000)	0.0059 (0.0000)	0.0004 (0.0000)

will follow the approach of Cowell and Flachaire (2007). For this define the relative impact of contamination as

$$RIC(I) = \frac{I(\hat{F}) - I(\hat{F}^*)}{I(\hat{F})}, \quad (17)$$

with small values (near zero) indicating insensitivity to contamination. We will contaminate the data set by multiplying a proportion $p = 1\%$ of the data by a constant $c = 10$. In order to avoid negative values, we use $|RIC(I)|$. The simulation results for the Gini coefficient are given in Table 2. Although we also obtained results for the other measures, we do not give them here in order not to clutter the main text of the paper. These results show that the semi-parametric estimators are substantially less sensitive to contaminations than the nonparametric estimators for all the underlying distributions.

Table 2: Relative impact of contamination (RIC) for Gini measure.

Gini (n=500)					Gini (n=1000)				
	NP	SPGPD	SPPa	SPPPD		NP	SPGPD	SPPa	SPPPD
Pa	0.0694 (0.004)	0.0603 (0.0156)	0.0606 (0.0092)	0.0601 (0.0024)	Pa	0.0729 (0.0038)	0.0478 (0.0056)	0.0401 (0.002)	0.0480 (0.0019)
Bu1	0.0172 (0.0016)	0.0150 (0.0106)	0.0156 (0.0639)	0.0152 (0.0009)	Bu1	0.0868 (0.0013)	0.0494 (0.0047)	0.0471 (0.009)	0.0427 (0.0008)
Bu2	0.0250 (0.0022)	0.0133 (0.0099)	0.0227 (0.0251)	0.0280 (0.0012)	Bu2	0.0726 (0.0017)	0.0278 (0.0051)	0.0291 (0.0057)	0.0259 (0.001)
Bu3	0.1050 (0.0038)	0.0669 (0.0094)	0.0677 (0.0017)	0.0644 (0.0013)	Bu3	0.1064 (0.0033)	0.0560 (0.0039)	0.0626 (0.0012)	0.0610 (0.0007)
Fr1	0.0661 (0.0036)	0.0441 (0.006)	0.0363 (0.0022)	0.0312 (0.0017)	Fr1	0.0676 (0.003)	0.0481 (0.0119)	0.0325 (0.001)	0.0349 (0.001)
Fr2	0.0702 (0.0035)	0.0386 (0.0079)	0.0374 (0.003)	0.0386 (0.0016)	Fr2	0.0654 (0.0029)	0.0435 (0.0065)	0.0327 (0.0012)	0.0349 (0.0011)
T2	0.0457 (0.0024)	0.0364 (0.0075)	0.0381 (0.0012)	0.0353 (0.0008)	T2	0.0480 (0.0019)	0.0256 (0.0027)	0.0205 (0.0006)	0.0274 (0.0006)
Average	0.0569 (0.0012)	0.0392 (0.0038)	0.0398 (0.0099)	0.0390 (0.0006)	Average	0.0742 (0.001)	0.0426 (0.0024)	0.0378 (0.0016)	0.0393 (0.0004)

Gini (n=5000)					Gini (n=10000)				
	NP	SPGPD	SPPa	SPPPD		NP	SPGPD	SPPa	SPPPD
Pa	0.0706 (0.0024)	0.0400 (0.0034)	0.0364 (0.0009)	0.0392 (0.0006)	Pa	0.0704 (0.002)	0.0360 (0.0005)	0.0379 (0.0007)	0.0382 (0.0004)
Bu1	0.0367 (0.0009)	0.0098 (0.0018)	0.0142 (0.0022)	0.0188 (0.0005)	Bu1	0.0199 (0.0008)	0.0183 (0.0003)	0.0192 (0.0015)	0.0193 (0.0003)
Bu2	0.0252 (0.0014)	0.0201 (0.0042)	0.0278 (0.0019)	0.0212 (0.0005)	Bu2	0.0243 (0.001)	0.0211 (0.0043)	0.0204 (0.0012)	0.0199 (0.0003)
Bu3	0.1063 (0.0019)	0.0521 (0.0006)	0.0635 (0.0006)	0.0594 (0.0003)	Bu3	0.1059 (0.0013)	0.0519 (0.0004)	0.0667 (0.0005)	0.0589 (0.0002)
Fr1	0.0655 (0.0017)	0.0377 (0.0065)	0.0393 (0.0005)	0.0313 (0.0004)	Fr1	0.0668 (0.0015)	0.0429 (0.0003)	0.0431 (0.0004)	0.0408 (0.0003)
Fr2	0.0674 (0.0017)	0.0347 (0.0018)	0.0397 (0.0006)	0.0315 (0.0004)	Fr2	0.0661 (0.0013)	0.0438 (0.0003)	0.0428 (0.0004)	0.0406 (0.0003)
T2	0.0489 (0.0012)	0.0228 (0.0004)	0.0211 (0.0004)	0.0235 (0.0002)	T2	0.0482 (0.0008)	0.0232 (0.0003)	0.0287 (0.0003)	0.0235 (0.0002)
Average	0.0601 (0.0006)	0.0310 (0.0013)	0.0346 (0.0005)	0.0321 (0.0002)	Average	0.0574 (0.0005)	0.0339 (0.0006)	0.0370 (0.0003)	0.0345 (0.0001)

4.3. Confidence intervals

The simulation results for confidence intervals in the case of the Pareto distribution are given in Table 3, where standard errors of the estimates are given in brackets. The values in bold indicate the better performance of the corresponding intervals in terms of CP and ACIL. Note that the perturbed Pareto distribution involves three parameters that need to satisfy a number of conditions to make the estimation possible. This requires using only simulated samples that lead to the satisfaction of such conditions. Handling this issue makes the simulation more complicated, especially since we make extensive use of the bootstrap. Thus we do not include the intervals based on the PPD in this paper. We will return to this issue in future work.

Throughout the simulation results we see that the confidence intervals based on semi-parametric estimators perform very well compared to the intervals based on the usual methods. In fact the

Table 3: Average confidence interval length (ACIL) and coverage probability (CP) for the measures of inequality when samples come from Pareto distribution.

Pareto (n = 1000)		NP		SPGPD		SPPa	
		BPI	BTI	BPI	BTI	BPI	BTI
Gini	ACIL	0.1604 (0.0191)	0.2551 (0.0345)	0.1661 (0.0172)	0.2941 (0.1211)	0.0987 (0.0126)	0.0714 (0.0094)
	CP	0.9306 (0.0224)	0.9271 (0.0124)	0.9615 (0.0109)	0.9227 (0.0155)	0.9475 (0.0204)	0.9321 (0.0245)
GE0	ACIL	0.3354 (0.0198)	0.7773 (0.0357)	0.2959 (0.0178)	0.5307 (0.1253)	0.5828 (0.013)	0.3988 (0.0097)
	CP	0.9614 (0.0221)	0.9480 (0.0122)	0.9483 (0.0107)	0.9395 (0.0153)	0.9552 (0.0201)	0.9589 (0.0241)
GE1	ACIL	0.8749 (0.0189)	0.8944 (0.0321)	0.3827 (0.0156)	0.6175 (0.1096)	0.6696 (0.012)	0.4856 (0.0089)
	CP	0.9420 (0.0252)	0.9492 (0.014)	0.9487 (0.0123)	0.9299 (0.0174)	0.9516 (0.023)	0.9493 (0.0276)
GE1.3	ACIL	0.9254 (0.0239)	0.8234 (0.0407)	0.3923 (0.0197)	0.6026 (0.1387)	0.6292 (0.0152)	0.4372 (0.0113)
	CP	0.9317 (0.0331)	0.9354 (0.0183)	0.9357 (0.0161)	0.9399 (0.0229)	0.9557 (0.0301)	0.9491 (0.0362)
QSR	ACIL	5.8249 (0.1253)	33.4915 (0.0097)	6.1707 (0.0321)	6.3172 (0.0156)	6.0743 (0.1096)	6.2185 (0.012)
	CP	0.9680 (0.0532)	0.9558 (0.0294)	0.9342 (0.0259)	0.9310 (0.0368)	0.9542 (0.0484)	0.9434 (0.0581)
Pareto (n = 5000)		NP		SPGPD		SPPa	
		BPI	BTI	BPI	BTI	BPI	BTI
Gini	ACIL	0.1022 (0.0112)	0.1786 (0.0257)	0.0824 (0.009)	0.2261 (0.0464)	0.0325 (0.0039)	0.0270 (0.0033)
	CP	0.9394 (0.0154)	0.9199 (0.0123)	0.9520 (0.0261)	0.9456 (0.0129)	0.9351 (0.0146)	0.9516 (0.0115)
GE0	ACIL	0.2124 (0.0116)	0.5689 (0.0266)	0.1524 (0.0093)	0.2459 (0.048)	0.3172 (0.004)	0.2025 (0.0034)
	CP	0.9607 (0.0152)	0.9488 (0.0121)	0.9600 (0.0257)	0.9540 (0.0137)	0.9641 (0.0144)	0.9594 (0.0113)
GE1	ACIL	0.8314 (0.0111)	0.9185 (0.0239)	0.2234 (0.0081)	0.3387 (0.042)	0.4143 (0.0037)	0.3061 (0.0031)
	CP	0.9143 (0.0173)	0.9262 (0.0138)	0.9504 (0.0294)	0.9521 (0.0145)	0.9511 (0.0164)	0.9498 (0.0129)
GE1.3	ACIL	0.8115 (0.014)	0.9225 (0.0303)	0.2392 (0.0103)	0.3432 (0.0532)	0.4047 (0.0047)	0.2893 (0.004)
	CP	0.9432 (0.0227)	0.9360 (0.0182)	0.9464 (0.0385)	0.9544 (0.0205)	0.9545 (0.0216)	0.9457 (0.0170)
QSR	ACIL	3.6624 (0.048)	12.8283 (0.0034)	3.1184 (0.0239)	3.1460 (0.0081)	3.0697 (0.042)	3.0968 (0.0037)
	CP	0.9726 (0.0365)	0.9473 (0.0292)	0.9307 (0.0619)	0.9210 (0.0306)	0.9507 (0.0346)	0.9415 (0.0273)

semi-parametric methods yield reasonable confidence interval lengths with quite satisfactory coverage probabilities (more or less around the desired coverage of 95%). In cases where the coverage probability is not very satisfactory, we can see that it is still closer to the desired coverage for the semi-parametric methods than for the nonparametric methods for most of the measures. Given the fact that confidence intervals are obtained for complex measures and the complexity of the statistics underlying the bootstrap, the semi-parametric procedures do remarkably well. Furthermore, the fact that the standard errors are reasonably small shows that the estimation procedures are quite accurate and the conclusions drawn are reliable. Similar conclusions are reached when simulating samples from the Burr and the Fréchet distributions instead of the Pareto distribution.

5. Application

In this section we illustrate the application of the proposed techniques to a real data set. The data considered is a sample of size 5000 from the 2005 South African income and expenditure survey data. The survey was conducted to examine income and expenditure as well as poverty and inequality in South Africa. Households that were sampled took part in the survey for one month after which new subsamples of households started taking part in the survey at the beginning of each month. A main questionnaire consisting of five interview modules is administered by a fieldworker to a selected household. Each interview was conducted on five different visits. The main questionnaire required households to account for their acquisitions of goods and services. Note that although the original survey data came from a complex sampling procedure, we took a simple random sample from it. The data set we work with will therefore be considered as a simple random sample from the survey data, taken as the population. We will refer to it as IES data.

In order to select the threshold, the procedure described in Section 4 is followed. What we did was to apply the parametric distribution to $k = 5, 6, \dots, 125$ exceedances (or up to 25% of the data), and for each case we calculated the mean squared error. The k value considered is the one corresponding to the smallest mean squared error. For IES data, the procedure led us to use approximately 10% of the values for both the parameter estimation and the application of the parametric distribution. A further investigation is needed for better choices of threshold. Table 4 gives the estimates of the inequality measures considered, using IES data. The values in brackets are the standard errors of the corresponding estimates.

Since the semi-parametric estimators are more reliable in that they are less sensitive to outliers, inference based on them leads to more reliable conclusions. In order to decide which of the three parametric distributions should be considered, we apply the measure of representativeness given in Section 3.4. to IES data. For this we calculated $R(IESdata, GPD)$, $R(IESdata, Pa)$ and $R(IESdata, PPD)$, and the largest value determines the preferred distribution to use. Applying the measure to 10% of the upper values of the data set considered lead to $R(IESdata, GPD) = 0.96404$, $R(IESdata, Pa) = 0.99657$ and $R(IESdata, PPD) = 0.99485$, showing that we would prefer the strict Pareto distribution for IES data, being marginally better than PPD. Clearly in this case the GPD is a much less preferred choice.

Based on both the nonparametric and semi-parametric estimators, we constructed 95% bootstrap percentile and bootstrap t intervals for the measures using IESdata (see e.g. Kpanzou (2011))

Table 4: Nonparametric and semi-parametric estimates of inequality measures.

IESdata	NP	SPGPD	SPPa	SPPPD
Gini	0.6990 (0.0113)	0.7132 (0.0117)	0.6838 (0.0135)	0.6860 (0.0167)
GE0	0.9661 (0.0202)	1.0122 (0.0135)	0.9621 (0.0127)	0.9636 (0.0174)
GE1	1.1531 (0.0213)	1.5312 (0.0229)	1.1294 (0.0202)	1.1279 (0.0192)
GE1.3	1.5007 (0.024)	1.6626 (0.0174)	1.4349 (0.0199)	1.4410 (0.0227)
QSR	40.6089 (0.0302)	42.4122 (0.02715)	41.8292 (0.0211)	41.9375 (0.0361)

for a description of the bootstrap procedures, and Davidson and Flachaire (2007) for more details on bootstrap inference for inequality measures). The coverage probabilities (CP) and average confidence interval lengths (ACIL) are given in Table 5 below, where BPI denotes the bootstrap percentile interval and BTI the bootstrap t interval. The values in brackets are the standard errors of the corresponding estimates. We give the results based on the preferred choice of parametric distribution, viz. Pareto. The results for the GPD are also given for reference purposes. All the results are based on 1000 bootstrap samples.

We can see from the table that the SP method using the strict Pareto performs better than the nonparametric method in that the coverage probabilities are closer to the desired coverage, whilst maintaining a competitive ACIL. Comparing the SPGPD to the NP method, we note a mixed set of results with SPGPD performing better in some cases and worse in other cases. Finally, note that the standard errors are small, showing that the methods applied seem to be quite accurate.

6. Conclusions and further research

In this paper we further developed the semi-parametric estimator of Cowell and Flachaire (2007) by explicitly employing results from extreme value theory in order to construct more reliable distribution estimators. These new estimators of the distribution formed the basis for less sensitive estimators of the measures of inequality. We focused, in particular, on estimating the Gini, the generalised entropy and the quintile share ratio measures and developing confidence intervals for them. From a simulation study we concluded that these semi-parametric estimators of the inequality measures are less sensitive to data contamination than their nonparametric counterparts. The mean squared errors are very reasonable except for small sample sizes – their performance was good for the Gini but less so for some of the other measures. Confidence intervals based on them, using the bootstrap, performed very well. The proposed methods therefore provide the users with more reliable possibilities than nonparametric methods in drawing inferences on inequality measures. As an illustration, we applied the methods to the IES data, estimating the inequality measures and constructing bootstrap confidence intervals based on both nonparametric and semi-parametric estimators. For this data set

Table 5: Confidence intervals for inequality measures.

Gini for IESdata	NP		SPPa		SPGPD	
	BPI	BTI	BPI	BTI	BPI	BTI
ACIL	0.0368 (0.0041)	0.0325 (0.0036)	0.0323 (0.0041)	0.0371 (0.0036)	0.0239 (0.0026)	0.0419 (0.0047)
CP	0.9211 (0.0036)	0.9097 (0.0038)	0.9509 (0.0031)	0.9497 (0.0029)	0.9357 (0.0033)	0.9336 (0.0037)
GE0 for IESdata	NP		SPPa		SPGPD	
	BPI	BTI	BPI	BTI	BPI	BTI
ACIL	0.0435 (0.0042)	0.0442 (0.0054)	0.0519 (0.0049)	0.0437 (0.0041)	0.0459 (0.0042)	0.0506 (0.0051)
CP	0.9424 (0.0026)	0.9147 (0.0022)	0.9515 (0.0019)	0.9448 (0.0018)	0.9383 (0.0025)	0.9432 (0.0025)
GE1 for IESdata	NP		SPPa		SPGPD	
	BPI	BTI	BPI	BTI	BPI	BTI
ACIL	0.0491 (0.0036)	0.0414 (0.0032)	0.0402 (0.0032)	0.0424 (0.0028)	0.0414 (0.0029)	0.039 (0.0035)
CP	0.9343 (0.0021)	0.8991 (0.0037)	0.9528 (0.0020)	0.9452 (0.0019)	0.9354 (0.0026)	0.9277 (0.0026)
GE1.3 for IESdata	NP		SPPa		SPGPD	
	BPI	BTI	BPI	BTI	BPI	BTI
ACIL	0.0431 (0.0042)	0.0438 (0.0054)	0.0514 (0.0049)	0.0433 (0.0041)	0.0455 (0.0042)	0.0501 (0.0051)
CP	0.9332 (0.0026)	0.9057 (0.0022)	0.9462 (0.0020)	0.9454 (0.0019)	0.9289 (0.0026)	0.9312 (0.0026)
QSR for IESdata	NP		SPPa		SPGPD	
	BPI	BTI	BPI	BTI	BPI	BTI
ACIL	0.0382 (0.0038)	0.0337 (0.0034)	0.0335 (0.0038)	0.0385 (0.0034)	0.0239 (0.0024)	0.0435 (0.0044)
CP	0.9471 (0.0038)	0.9354 (0.0034)	0.9513 (0.0041)	0.9429 (0.0023)	0.9409 (0.0027)	0.9358 (0.0035)

the semi-parametric procedures proved to perform better than the nonparametric ones in most of the cases.

Our further research on these semi-parametric estimators includes studying their asymptotic distribution theory and modifying them in order to apply to complex sampling schemes. Furthermore, they will also be compared to using a mixture model for fitting a distribution to heavy tails. All these aspects are currently under investigation and will be reported on in future.

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